A Computational Approach to Identifying Evolutionarily Stable Strategies in Mammalian Search Behaviour

Supplementary Material B: Sampling Intervals

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Effect of the sampling frequency

The sampling frequency was calculated as the the rounded velocity autocorrelation (τ_v) divided by factor x. 200 values of x from 0.5 to 100 were sampled. We used a body mass of 105.5kg (105500g) on a landscape with 500 patches, no clustering, and no caloric variance. We then generated three movement paths (using seeds 1, 2, and 3), from the same movement model, to ensure consistency of the patch visits.

To determine a reasonable sampling interval, we fit a Generalized Additive Model (GAM) to the data and calculated the x necessary to reach 95% of the maximum patch visits recorded. The results of this analysis showed inconsistent results directly at the 95% interval, such that a sampling interval of approximately 38 seconds was choosen to be well ahead of the 95% mark. This corresponds with a x value of 10, thus the sampling interval used for subsequent tests was determined to be $\frac{1}{10}\tau_v$. The results of this GAM model were used to determine the sampling interval. Figure 1 shows the results of a GAM model created with the `geom_smooth()` function from the `ggplot2` package.

In addition, we have included the three movement tracks used to generate this data, overlaid on the full landscape used, in Figure 2.

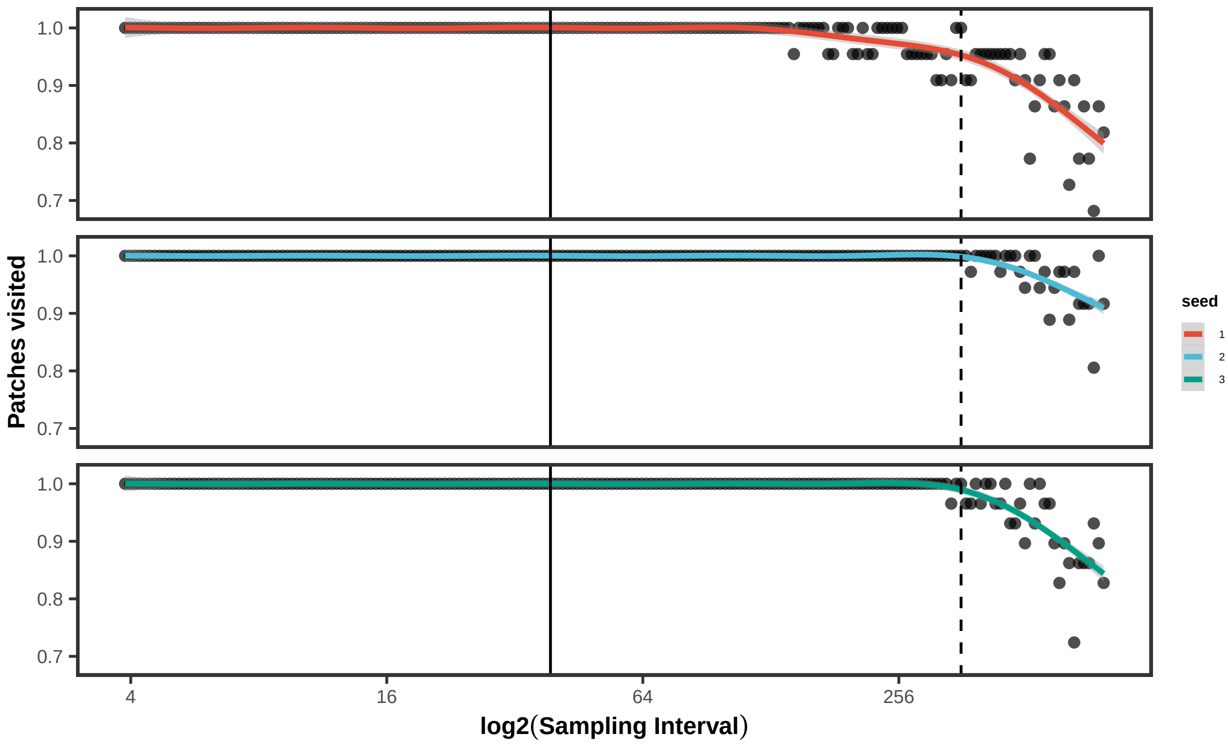

Figure 1: The proportion of patch visits detected as a function of sampling interval chosen, grouped by their seed. The coloured lines show the estimated relationship based on a Generalized Additive Model fit using the `geom_smooth()` function from the `ggplot2` package. The solid vertical line shows sampling interval of 38 seconds, corresponding to x equal to 10. The dashed vertical line shows the interval to reach 95% of encounters across seeds.

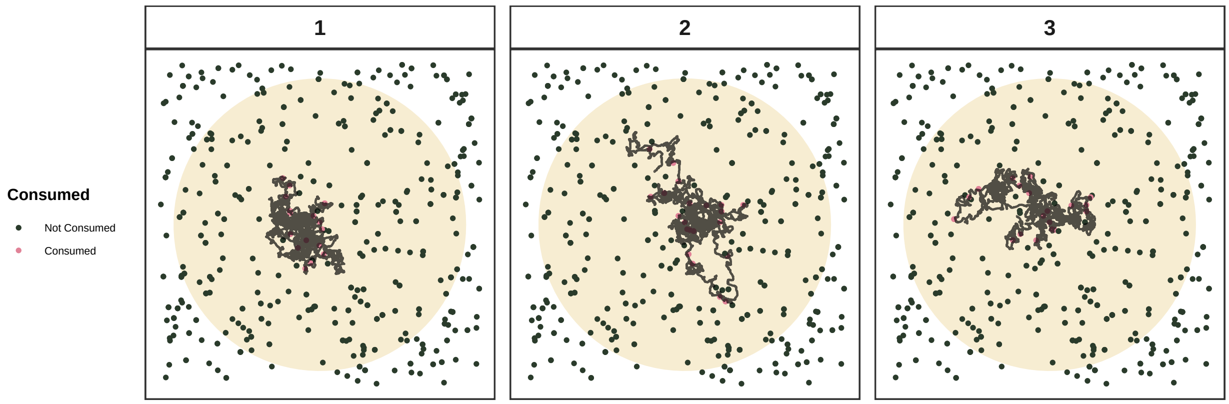

Figure 2: Full movement tracks and consumed patches for each seed of the analysis, prior to being thinned. Patches are coloured by their consumption status. The yellow circle represents the 95% home range area. Tracks are labelled by their seed.